



June 16, 2025

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: UCMR5_SE2_May2025
Pace Project No.: 35956761

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on May 20, 2025. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Ormond Beach

If you have any questions concerning this report, please feel free to contact me.

Sincerely,

A handwritten signature in black ink that reads 'Rebecca Selph'.

Rebecca Selph
rebecca.selph@pacelabs.com
(386)672-5668
Project Manager

Enclosures



REPORT OF LABORATORY ANALYSIS

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CERTIFICATIONS

Project: UCMR5_SE2_May2025

Pace Project No.: 35956761

Pace Analytical Services Ormond Beach

8 East Tower Circle, Ormond Beach, FL 32174

Alaska DEC- CS/UST/LUST

Alabama Certification #: 41320

California Certification# 3096

Colorado Certification: FL NELAC Reciprocity

Connecticut Certification #: PH-0216

Delaware Certification: FL NELAC Reciprocity

DoD-ANAB #:ADE-3199

Florida Certification #: E83079

Georgia Certification #: 955

Guam Certification: FL NELAC Reciprocity

Hawaii Certification: FL NELAC Reciprocity

Illinois Certification #: 200068

Indiana Certification: FL NELAC Reciprocity

Kansas Certification #: E-10383

Kentucky Certification #: 90050

Louisiana Certification #: FL NELAC Reciprocity

Louisiana Environmental Certificate #: 05007

Maine Certification #: FL01264

Maryland Certification: #346

Massachusetts Certification #: M-FL1264

Michigan Certification #: 9911

Mississippi Certification: FL NELAC Reciprocity

Missouri Certification #: 236

Montana Certification #: Cert 0074

Nebraska Certification: NE-OS-28-14

Nevada Certification: FL NELAC Reciprocity

New Hampshire Certification #: 2958

New Jersey Certification #: FL022

New York Certification #: 11608

North Carolina Environmental Certificate #: 667

North Carolina Certification #: 12710

North Dakota Certification #: R-216

Ohio DEP 87780

Oklahoma Certification #: D9947

Pennsylvania Certification #: 68-00547

Puerto Rico Certification #: FL01264

South Carolina Certification: #96042001

Tennessee Certification #: TN02974

Texas Certification: FL NELAC Reciprocity

US Virgin Islands Certification: FL NELAC Reciprocity

Utah

Utah FL NELAC Reciprocity

Virginia Environmental Certification #: 460165

West Virginia Certification #: 9962C

Wisconsin Certification #: 399079670

Wyoming (EPA Region 8): FL NELAC Reciprocity

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SAMPLE SUMMARY

Project: UCMR5_SE2_May2025

Pace Project No.: 35956761

Lab ID	Sample ID	Matrix	Date Collected	Date Received
35956761001	10413 Crosby Wellfield Service	Drinking Water	05/19/25 10:40	05/20/25 11:00
35956761002	FRB-10413 Crosby Wellfield Ser	Drinking Water	05/19/25 10:38	05/20/25 11:00
35956761003	12269 44 Road Water Service Bu	Drinking Water	05/19/25 10:05	05/20/25 11:00
35956761004	FRB-12269 44 Road Water Servic	Drinking Water	05/19/25 10:03	05/20/25 11:00

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SAMPLE ANALYTE COUNT

Project: UCMR5_SE2_May2025

Pace Project No.: 35956761

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory	
35956761001	10413	Crosby Wellfield Service	EPA 200.7	ASB	1	PASI-O
			EPA 533	SWR	41	PASI-O
			EPA 537.1	SWR	8	PASI-O
35956761003	12269	44 Road Water Service Bu	EPA 200.7	ASB	1	PASI-O
			EPA 533	SWR	41	PASI-O
			EPA 537.1	SWR	8	PASI-O

PASI-O = Pace Analytical Services - Ormond Beach

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ANALYTICAL RESULTS

Project: UCMR5_SE2_May2025

Pace Project No.: 35956761

Sample: 10413 Crosby Wellfield Lab ID: 35956761001 Collected: 05/19/25 10:40 Received: 05/20/25 11:00 Matrix: Drinking Water
Service

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
200.7 MET ICP, UCMR									
Analytical Method: EPA 200.7 Preparation Method: EPA 200.7									
Initial Volume/Weight: 50 mL Final Volume/Weight: 50 mL									
Pace Analytical Services - Ormond Beach									
Lithium	7.5 U	ug/L	22.5	7.5	2.5	06/03/25 04:18	06/14/25 11:35	7439-93-2	N2
533 PFAS Compounds, UCMR									
Analytical Method: EPA 533 Preparation Method: EPA 533									
Initial Volume/Weight: 274.79 mL Final Volume/Weight: 1 mL									
Pace Analytical Services - Ormond Beach									
11CI-PF3OUdS	0.0015 U	ug/L	0.0045	0.0015	1	05/22/25 10:34	05/23/25 21:18	763051-92-9	
4:2 FTS	0.0009 U	ug/L	0.0027	0.0009	1	05/22/25 10:34	05/23/25 21:18	757124-72-4	
6:2 FTS	0.0015 U	ug/L	0.0045	0.0015	1	05/22/25 10:34	05/23/25 21:18	27619-97-2	
8:2 FTS	0.0015 U	ug/L	0.0045	0.0015	1	05/22/25 10:34	05/23/25 21:18	39108-34-4	
9CI-PF3ONS	0.0006 U	ug/L	0.0018	0.0006	1	05/22/25 10:34	05/23/25 21:18	756426-58-1	
ADONA	0.0009 U	ug/L	0.0027	0.0009	1	05/22/25 10:34	05/23/25 21:18	919005-14-4	
HFPO-DA	0.0015 U	ug/L	0.0045	0.0015	1	05/22/25 10:34	05/23/25 21:18	13252-13-6	
NFDHA	0.0061 U	ug/L	0.018	0.0061	1	05/22/25 10:34	05/23/25 21:18	151772-58-6	
PFBS	0.0009 U	ug/L	0.0027	0.0009	1	05/22/25 10:34	05/23/25 21:18	375-73-5	
PFDA	0.0009 U	ug/L	0.0027	0.0009	1	05/22/25 10:34	05/23/25 21:18	335-76-2	
PFHxA	0.0009 U	ug/L	0.0027	0.0009	1	05/22/25 10:34	05/23/25 21:18	307-24-4	
PFBA	0.0015 U	ug/L	0.0045	0.0015	1	05/22/25 10:34	05/23/25 21:18	375-22-4	
PFEESA	0.0009 U	ug/L	0.0027	0.0009	1	05/22/25 10:34	05/23/25 21:18	113507-82-7	
PFHpS	0.0009 U	ug/L	0.0027	0.0009	1	05/22/25 10:34	05/23/25 21:18	375-92-8	
PFMBA	0.0009 U	ug/L	0.0027	0.0009	1	05/22/25 10:34	05/23/25 21:18	863090-89-5	
PFMPA	0.0012 U	ug/L	0.0036	0.0012	1	05/22/25 10:34	05/23/25 21:18	377-73-1	
PFPeA	0.0009 U	ug/L	0.0027	0.0009	1	05/22/25 10:34	05/23/25 21:18	2706-90-3	
PFPeS	0.0012 U	ug/L	0.0036	0.0012	1	05/22/25 10:34	05/23/25 21:18	2706-91-4	
PFDoA	0.0009 U	ug/L	0.0027	0.0009	1	05/22/25 10:34	05/23/25 21:18	307-55-1	
PFHpA	0.0009 U	ug/L	0.0027	0.0009	1	05/22/25 10:34	05/23/25 21:18	375-85-9	
PFHxS	0.0009 U	ug/L	0.0027	0.0009	1	05/22/25 10:34	05/23/25 21:18	355-46-4	
PFNA	0.0012 U	ug/L	0.0036	0.0012	1	05/22/25 10:34	05/23/25 21:18	375-95-1	
PFOS	0.0012 U	ug/L	0.0036	0.0012	1	05/22/25 10:34	05/23/25 21:18	1763-23-1	
PFOA	0.0012 U	ug/L	0.0036	0.0012	1	05/22/25 10:34	05/23/25 21:18	335-67-1	
PFUnA	0.0006 U	ug/L	0.0018	0.0006	1	05/22/25 10:34	05/23/25 21:18	2058-94-8	
Surrogates									
13C24:2FTS (S)	105	%	50-200		1	05/22/25 10:34	05/23/25 21:18		
13C26:2FTS (S)	89	%	50-200		1	05/22/25 10:34	05/23/25 21:18		
13C28:2FTS (S)	90	%	50-200		1	05/22/25 10:34	05/23/25 21:18		
13C2-PFDoA (S)	78	%	50-200		1	05/22/25 10:34	05/23/25 21:18		
13C3HFPO-DA(S)	93	%	50-200		1	05/22/25 10:34	05/23/25 21:18		
13C3-PFBS (S)	95	%	50-200		1	05/22/25 10:34	05/23/25 21:18		
13C3-PFHxS (S)	95	%	50-200		1	05/22/25 10:34	05/23/25 21:18		
13C4-PFBA (S)	95	%	50-200		1	05/22/25 10:34	05/23/25 21:18		
13C4-PFHpA (S)	90	%	50-200		1	05/22/25 10:34	05/23/25 21:18		
13C5-PFHxA (S)	93	%	50-200		1	05/22/25 10:34	05/23/25 21:18		
13C5-PFPeA (S)	93	%	50-200		1	05/22/25 10:34	05/23/25 21:18		
13C6-PFDA (S)	84	%	50-200		1	05/22/25 10:34	05/23/25 21:18		

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: UCMR5_SE2_May2025

Pace Project No.: 35956761

Sample: **10413 Crosby Wellfield Service** Lab ID: **35956761001** Collected: 05/19/25 10:40 Received: 05/20/25 11:00 Matrix: Drinking Water

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
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533 PFAS Compounds, UCMR

Analytical Method: EPA 533 Preparation Method: EPA 533
Initial Volume/Weight: 274.79 mL Final Volume/Weight: 1 mL
Pace Analytical Services - Ormond Beach

Surrogates

13C7-PFUDA (S)	83	%	50-200		1	05/22/25 10:34	05/23/25 21:18		
13C8-PFOA (S)	94	%	50-200		1	05/22/25 10:34	05/23/25 21:18		
13C8-PFOS (S)	94	%	50-200		1	05/22/25 10:34	05/23/25 21:18		
13C9-PFNA (S)	89	%	50-200		1	05/22/25 10:34	05/23/25 21:18		

537.1 PFAS Compounds, UCMR

Analytical Method: EPA 537.1 Preparation Method: EPA 537.1
Initial Volume/Weight: 274.44 mL Final Volume/Weight: 1 mL
Pace Analytical Services - Ormond Beach

NEtFOSAA	0.0015 U	ug/L	0.0055	0.0015	1	05/22/25 08:58	06/13/25 19:39	2991-50-6	
NMeFOSAA	0.0018 U	ug/L	0.0055	0.0018	1	05/22/25 08:58	06/13/25 19:39	2355-31-9	
PFTeDA	0.0024 U	ug/L	0.0073	0.0024	1	05/22/25 08:58	06/13/25 19:39	376-06-7	
PFTrDA	0.0021 U	ug/L	0.0064	0.0021	1	05/22/25 08:58	06/13/25 19:39	72629-94-8	

Surrogates

13C2-PFDA (S)	116	%	70-130		1	05/22/25 08:58	06/13/25 19:39		
13C2-PFHxA (S)	110	%	70-130		1	05/22/25 08:58	06/13/25 19:39		
NEtFOSAA-d5 (S)	114	%	70-130		1	05/22/25 08:58	06/13/25 19:39		
HFPO-DAS (S)	104	%	70-130		1	05/22/25 08:58	06/13/25 19:39		

Sample: **12269 44 Road Water Service Bu** Lab ID: **35956761003** Collected: 05/19/25 10:05 Received: 05/20/25 11:00 Matrix: Drinking Water

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
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200.7 MET ICP, UCMR

Analytical Method: EPA 200.7 Preparation Method: EPA 200.7
Initial Volume/Weight: 50 mL Final Volume/Weight: 50 mL
Pace Analytical Services - Ormond Beach

Lithium	7.5 U	ug/L	22.5	7.5	2.5	06/03/25 04:18	06/14/25 11:37	7439-93-2	N2
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533 PFAS Compounds, UCMR

Analytical Method: EPA 533 Preparation Method: EPA 533
Initial Volume/Weight: 274.64 mL Final Volume/Weight: 1 mL
Pace Analytical Services - Ormond Beach

11CI-PF3OUdS	0.0015 U	ug/L	0.0046	0.0015	1	05/22/25 10:34	05/23/25 21:35	763051-92-9	
4:2 FTS	0.0009 U	ug/L	0.0027	0.0009	1	05/22/25 10:34	05/23/25 21:35	757124-72-4	
6:2 FTS	0.0015 U	ug/L	0.0046	0.0015	1	05/22/25 10:34	05/23/25 21:35	27619-97-2	
8:2 FTS	0.0015 U	ug/L	0.0046	0.0015	1	05/22/25 10:34	05/23/25 21:35	39108-34-4	
9CI-PF3ONS	0.0006 U	ug/L	0.0018	0.0006	1	05/22/25 10:34	05/23/25 21:35	756426-58-1	
ADONA	0.0009 U	ug/L	0.0027	0.0009	1	05/22/25 10:34	05/23/25 21:35	919005-14-4	
HFPO-DA	0.0015 U	ug/L	0.0046	0.0015	1	05/22/25 10:34	05/23/25 21:35	13252-13-6	
NFDHA	0.0061 U	ug/L	0.018	0.0061	1	05/22/25 10:34	05/23/25 21:35	151772-58-6	
PFBS	0.0009 U	ug/L	0.0027	0.0009	1	05/22/25 10:34	05/23/25 21:35	375-73-5	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: UCMR5_SE2_May2025

Pace Project No.: 35956761

Sample: 12269 44 Road Water **Lab ID:** 35956761003 **Collected:** 05/19/25 10:05 **Received:** 05/20/25 11:00 **Matrix:** Drinking Water
Service Bu

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
533 PFAS Compounds, UCMR									
Analytical Method: EPA 533 Preparation Method: EPA 533									
Initial Volume/Weight: 274.64 mL Final Volume/Weight: 1 mL									
Pace Analytical Services - Ormond Beach									
PFDA	0.0009 U	ug/L	0.0027	0.0009	1	05/22/25 10:34	05/23/25 21:35	335-76-2	
PFHxA	0.0009 U	ug/L	0.0027	0.0009	1	05/22/25 10:34	05/23/25 21:35	307-24-4	
PFBA	0.0015 U	ug/L	0.0046	0.0015	1	05/22/25 10:34	05/23/25 21:35	375-22-4	
PFEESA	0.0009 U	ug/L	0.0027	0.0009	1	05/22/25 10:34	05/23/25 21:35	113507-82-7	
PFHpS	0.0009 U	ug/L	0.0027	0.0009	1	05/22/25 10:34	05/23/25 21:35	375-92-8	
PFMBA	0.0009 U	ug/L	0.0027	0.0009	1	05/22/25 10:34	05/23/25 21:35	863090-89-5	
PFMPA	0.0012 U	ug/L	0.0036	0.0012	1	05/22/25 10:34	05/23/25 21:35	377-73-1	
PFPeA	0.0009 U	ug/L	0.0027	0.0009	1	05/22/25 10:34	05/23/25 21:35	2706-90-3	
PFPeS	0.0012 U	ug/L	0.0036	0.0012	1	05/22/25 10:34	05/23/25 21:35	2706-91-4	
PFDaA	0.0009 U	ug/L	0.0027	0.0009	1	05/22/25 10:34	05/23/25 21:35	307-55-1	
PFHpA	0.0009 U	ug/L	0.0027	0.0009	1	05/22/25 10:34	05/23/25 21:35	375-85-9	
PFHxS	0.0009 U	ug/L	0.0027	0.0009	1	05/22/25 10:34	05/23/25 21:35	355-46-4	
PFNA	0.0012 U	ug/L	0.0036	0.0012	1	05/22/25 10:34	05/23/25 21:35	375-95-1	
PFOS	0.0012 U	ug/L	0.0036	0.0012	1	05/22/25 10:34	05/23/25 21:35	1763-23-1	
PFOA	0.0012 U	ug/L	0.0036	0.0012	1	05/22/25 10:34	05/23/25 21:35	335-67-1	
PFUnA	0.0006 U	ug/L	0.0018	0.0006	1	05/22/25 10:34	05/23/25 21:35	2058-94-8	
Surrogates									
13C24:2FTS (S)	104	%	50-200		1	05/22/25 10:34	05/23/25 21:35		
13C26:2FTS (S)	88	%	50-200		1	05/22/25 10:34	05/23/25 21:35		
13C28:2FTS (S)	88	%	50-200		1	05/22/25 10:34	05/23/25 21:35		
13C2-PFDaA (S)	82	%	50-200		1	05/22/25 10:34	05/23/25 21:35		
13C3HFPO-DA(S)	94	%	50-200		1	05/22/25 10:34	05/23/25 21:35		
13C3-PFBS (S)	95	%	50-200		1	05/22/25 10:34	05/23/25 21:35		
13C3-PFHxS (S)	94	%	50-200		1	05/22/25 10:34	05/23/25 21:35		
13C4-PFBA (S)	95	%	50-200		1	05/22/25 10:34	05/23/25 21:35		
13C4-PFHpA (S)	91	%	50-200		1	05/22/25 10:34	05/23/25 21:35		
13C5-PFHxA (S)	94	%	50-200		1	05/22/25 10:34	05/23/25 21:35		
13C5-PFPeA (S)	95	%	50-200		1	05/22/25 10:34	05/23/25 21:35		
13C6-PFDA (S)	84	%	50-200		1	05/22/25 10:34	05/23/25 21:35		
13C7-PFUdA (S)	85	%	50-200		1	05/22/25 10:34	05/23/25 21:35		
13C8-PFOA (S)	93	%	50-200		1	05/22/25 10:34	05/23/25 21:35		
13C8-PFOS (S)	93	%	50-200		1	05/22/25 10:34	05/23/25 21:35		
13C9-PFNA (S)	89	%	50-200		1	05/22/25 10:34	05/23/25 21:35		

537.1 PFAS Compounds, UCMR

Analytical Method: EPA 537.1 Preparation Method: EPA 537.1

Initial Volume/Weight: 261.68 mL Final Volume/Weight: 1 mL

Pace Analytical Services - Ormond Beach

NEtFOSAA	0.0016 U	ug/L	0.0057	0.0016	1	05/22/25 08:58	06/13/25 19:55	2991-50-6	
NMeFOSAA	0.0019 U	ug/L	0.0057	0.0019	1	05/22/25 08:58	06/13/25 19:55	2355-31-9	
PFTeDA	0.0026 U	ug/L	0.0076	0.0026	1	05/22/25 08:58	06/13/25 19:55	376-06-7	
PFTTrDA	0.0023 U	ug/L	0.0067	0.0023	1	05/22/25 08:58	06/13/25 19:55	72629-94-8	
Surrogates									
13C2-PFDA (S)	118	%	70-130		1	05/22/25 08:58	06/13/25 19:55		

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ANALYTICAL RESULTS

Project: UCMR5_SE2_May2025

Pace Project No.: 35956761

Sample: 12269 44 Road Water Lab ID: 35956761003 Collected: 05/19/25 10:05 Received: 05/20/25 11:00 Matrix: Drinking Water
Service Bu

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
537.1 PFAS Compounds, UCMR									
Analytical Method: EPA 537.1 Preparation Method: EPA 537.1									
Initial Volume/Weight: 261.68 mL Final Volume/Weight: 1 mL									
Pace Analytical Services - Ormond Beach									
Surrogates									
13C2-PFHxA (S)	115	%	70-130		1	05/22/25 08:58	06/13/25 19:55		
NEtFOSAA-d5 (S)	111	%	70-130		1	05/22/25 08:58	06/13/25 19:55		
HFPO-DAS (S)	108	%	70-130		1	05/22/25 08:58	06/13/25 19:55		

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QUALITY CONTROL DATA

Project: UCMR5_SE2_May2025

Pace Project No.: 35956761

QC Batch:	1102849	Analysis Method:	EPA 200.7
QC Batch Method:	EPA 200.7	Analysis Description:	200.7 MET Drinking Water
		Laboratory:	Pace Analytical Services - Ormond Beach

Associated Lab Samples: 35956761001, 35956761003

METHOD BLANK: 6049346 Matrix: Drinking Water

Associated Lab Samples: 35956761001, 35956761003

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Lithium	ug/L	7.5 U	22.5	7.5	06/14/25 10:48	N2

LABORATORY CONTROL SAMPLE: 6049347

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Lithium	ug/L	9	9.3 I	103	50-150	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 6049348 6049349

Parameter	Units	35956752001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Lithium	ug/L	7.5 U	9	9	9.7 I	9.6 I	101	99	0-200		30	N2

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

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QUALITY CONTROL DATA

Project: UCMR5_SE2_May2025

Pace Project No.: 35956761

QC Batch: 1100351

Analysis Method: EPA 533

QC Batch Method: EPA 533

Analysis Description: 533 PFAS Compounds, UCMR

Laboratory: Pace Analytical Services - Ormond Beach

Associated Lab Samples: 35956761001, 35956761003

METHOD BLANK: 6036600

Matrix: Drinking Water

Associated Lab Samples: 35956761001, 35956761003

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
11CI-PF3OUdS	ug/L	0.0017 U	0.0050	0.0017	05/23/25 18:27	
4:2 FTS	ug/L	0.0010 U	0.0030	0.0010	05/23/25 18:27	
6:2 FTS	ug/L	0.0017 U	0.0050	0.0017	05/23/25 18:27	
8:2 FTS	ug/L	0.0017 U	0.0050	0.0017	05/23/25 18:27	
9CI-PF3ONS	ug/L	0.0007 U	0.0020	0.0007	05/23/25 18:27	
ADONA	ug/L	0.0010 U	0.0030	0.0010	05/23/25 18:27	
HFPO-DA	ug/L	0.0017 U	0.0050	0.0017	05/23/25 18:27	
NFDHA	ug/L	0.0067 U	0.020	0.0067	05/23/25 18:27	
PFBA	ug/L	0.0017 U	0.0050	0.0017	05/23/25 18:27	
PFBS	ug/L	0.0010 U	0.0030	0.0010	05/23/25 18:27	
PFDA	ug/L	0.0010 U	0.0030	0.0010	05/23/25 18:27	
PFDaA	ug/L	0.0010 U	0.0030	0.0010	05/23/25 18:27	
PFEESA	ug/L	0.0010 U	0.0030	0.0010	05/23/25 18:27	
PFHpA	ug/L	0.0010 U	0.0030	0.0010	05/23/25 18:27	
PFHpS	ug/L	0.0010 U	0.0030	0.0010	05/23/25 18:27	
PFHxA	ug/L	0.0010 U	0.0030	0.0010	05/23/25 18:27	
PFHxS	ug/L	0.0010 U	0.0030	0.0010	05/23/25 18:27	
PFMBA	ug/L	0.0010 U	0.0030	0.0010	05/23/25 18:27	
PFMPA	ug/L	0.0013 U	0.0040	0.0013	05/23/25 18:27	
PFNA	ug/L	0.0013 U	0.0040	0.0013	05/23/25 18:27	
PFOA	ug/L	0.0013 U	0.0040	0.0013	05/23/25 18:27	
PFOS	ug/L	0.0013 U	0.0040	0.0013	05/23/25 18:27	
PFPeA	ug/L	0.0010 U	0.0030	0.0010	05/23/25 18:27	
PFPeS	ug/L	0.0013 U	0.0040	0.0013	05/23/25 18:27	
PFUnA	ug/L	0.0007 U	0.0020	0.0007	05/23/25 18:27	
13C2-PFDaA (S)	%	82	50-200		05/23/25 18:27	
13C24:2FTS (S)	%	96	50-200		05/23/25 18:27	
13C26:2FTS (S)	%	90	50-200		05/23/25 18:27	
13C28:2FTS (S)	%	91	50-200		05/23/25 18:27	
13C3-PFBS (S)	%	93	50-200		05/23/25 18:27	
13C3-PFHxS (S)	%	92	50-200		05/23/25 18:27	
13C3HFPO-DA(S)	%	90	50-200		05/23/25 18:27	
13C4-PFBA (S)	%	88	50-200		05/23/25 18:27	
13C4-PFHpA (S)	%	87	50-200		05/23/25 18:27	
13C5-PFHxA (S)	%	90	50-200		05/23/25 18:27	
13C5-PFPeA (S)	%	90	50-200		05/23/25 18:27	
13C6-PFDA (S)	%	85	50-200		05/23/25 18:27	
13C7-PFUDa (S)	%	86	50-200		05/23/25 18:27	
13C8-PFOA (S)	%	91	50-200		05/23/25 18:27	
13C8-PFOS (S)	%	92	50-200		05/23/25 18:27	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: UCMR5_SE2_May2025

Pace Project No.: 35956761

METHOD BLANK: 6036600

Matrix: Drinking Water

Associated Lab Samples: 35956761001, 35956761003

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
13C9-PFNA (S)	%	89	50-200		05/23/25 18:27	

LABORATORY CONTROL SAMPLE: 6036601

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
11CI-PF3OUdS	ug/L	0.0019	0.0017 I	89	50-150	
4:2 FTS	ug/L	0.0019	0.0019 I	101	50-150	
6:2 FTS	ug/L	0.0019	0.0018 I	97	50-150	
8:2 FTS	ug/L	0.0019	0.0019 I	101	50-150	
9CI-PF3ONS	ug/L	0.0019	0.0018 I	95	50-150	
ADONA	ug/L	0.0019	0.0018 I	95	50-150	
HFPO-DA	ug/L	0.002	0.0019 I	93	50-150	
NFDHA	ug/L	0.02	0.022	108	50-150	
PFBA	ug/L	0.002	0.0020 I	98	50-150	
PFBS	ug/L	0.0018	0.0017 I	95	50-150	
PFDA	ug/L	0.002	0.0020 I	100	50-150	
PFDaA	ug/L	0.002	0.0020 I	101	50-150	
PFEESA	ug/L	0.0018	0.0017 I	95	50-150	
PFHpA	ug/L	0.002	0.0020 I	100	50-150	
PFHpS	ug/L	0.0019	0.0020 I	103	50-150	
PFHxA	ug/L	0.002	0.0019 I	97	50-150	
PFHxS	ug/L	0.0018	0.0020 I	108	50-150	
PFMBA	ug/L	0.002	0.0019 I	97	50-150	
PFMPA	ug/L	0.002	0.0020 I	101	50-150	
PFNA	ug/L	0.002	0.0020 I	100	50-150	
PFOA	ug/L	0.002	0.0020 I	101	50-150	
PFOS	ug/L	0.0019	0.0018 I	96	50-150	
PFPeA	ug/L	0.002	0.0021 I	105	50-150	
PFPeS	ug/L	0.0019	0.0019 I	103	50-150	
PFUnA	ug/L	0.002	0.0020 I	98	50-150	
13C2-PFDaA (S)	%			80	50-200	
13C24:2FTS (S)	%			98	50-200	
13C26:2FTS (S)	%			96	50-200	
13C28:2FTS (S)	%			92	50-200	
13C3-PFBS (S)	%			96	50-200	
13C3-PFHxS (S)	%			94	50-200	
13C3HFPO-DA(S)	%			92	50-200	
13C4-PFBA (S)	%			88	50-200	
13C4-PFHpA (S)	%			87	50-200	
13C5-PFHxA (S)	%			89	50-200	
13C5-PFPeA (S)	%			90	50-200	
13C6-PFDA (S)	%			84	50-200	
13C7-PFUdA (S)	%			85	50-200	
13C8-PFOA (S)	%			91	50-200	

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QUALITY CONTROL DATA

Project: UCMR5_SE2_May2025

Pace Project No.: 35956761

LABORATORY CONTROL SAMPLE: 6036601

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
13C8-PFOS (S)	%			93	50-200	
13C9-PFNA (S)	%			87	50-200	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 6036602 6036603

Parameter	Units	35956752001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
11CI-PF3OUdS	ug/L	0.0015 U	0.0018	0.0018	0.0017 I	0.0017 I	96	93	70-130		30	
4:2 FTS	ug/L	0.0009 U	0.0018	0.0018	0.0018 I	0.0017 I	98	95	70-130		30	
6:2 FTS	ug/L	0.0015 U	0.0018	0.0018	0.0018 I	0.0019 I	95	100	70-130		30	
8:2 FTS	ug/L	0.0015 U	0.0018	0.0018	0.0019 I	0.0017 I	107	95	70-130		30	
9CI-PF3ONS	ug/L	0.0006 U	0.0018	0.0018	0.0018 I	0.0017 I	98	95	70-130		30	
ADONA	ug/L	0.0009 U	0.0018	0.0018	0.0017 I	0.0018 I	97	101	70-130		30	
HFPO-DA	ug/L	0.0015 U	0.0019	0.0019	0.0020 I	0.0020 I	107	108	70-130		30	
NFDHA	ug/L	0.0060 U	0.019	0.019	0.017 I	0.018 I	90	91	70-130		30	
PFBA	ug/L	0.0015 U	0.0019	0.0019	0.0026 I	0.0026 I	98	100	70-130		30	
PFBS	ug/L	0.0009 U	0.0017	0.0017	0.0019 I	0.0017 I	97	83	70-130		30	
PFDA	ug/L	0.0009 U	0.0019	0.0019	0.0019 I	0.0019 I	101	99	70-130		30	
PFDaA	ug/L	0.0009 U	0.0019	0.0019	0.0018 I	0.0019 I	96	99	70-130		30	
PFEESA	ug/L	0.0009 U	0.0017	0.0017	0.0016 I	0.0016 I	97	97	70-130		30	
PFHpA	ug/L	0.0009 U	0.0019	0.0019	0.0021 I	0.0020 I	103	99	70-130		30	
PFHpS	ug/L	0.0009 U	0.0018	0.0018	0.0019 I	0.0018 I	106	102	70-130		30	
PFHxA	ug/L	0.0009 U	0.0019	0.0019	0.0022 I	0.0021 I	102	102	70-130		30	
PFHxS	ug/L	0.0009 U	0.0017	0.0017	0.0020 I	0.0018 I	108	99	70-130		30	
PFMBA	ug/L	0.0009 U	0.0019	0.0019	0.0019 I	0.0019 I	100	99	70-130		30	
PFMPA	ug/L	0.0012 U	0.0019	0.0019	0.0019 I	0.0019 I	101	102	70-130		30	
PFNA	ug/L	0.0012 U	0.0019	0.0019	0.0020 I	0.0021 I	98	104	70-130		30	
PFOA	ug/L	0.0012 U	0.0019	0.0019	0.0023 I	0.0022 I	104	96	70-130		30	
PFOS	ug/L	0.0012 U	0.0018	0.0018	0.0023 I	0.0023 I	96	96	70-130		30	
PFPeA	ug/L	0.0009 U	0.0019	0.0019	0.0023 I	0.0021 I	100	93	70-130		30	
PFPeS	ug/L	0.0012 U	0.0018	0.0018	0.0016 I	0.0015 I	90	83	70-130		30	
PFUnA	ug/L	0.0006 U	0.0019	0.0019	0.0018 I	0.0018 I	95	96	70-130		30	
13C2-PFDaA (S)	%						81	79	50-200			
13C24:2FTS (S)	%						142	148	50-200			
13C26:2FTS (S)	%						104	105	50-200			
13C28:2FTS (S)	%						96	99	50-200			
13C3-PFBS (S)	%						90	91	50-200			
13C3-PFHxS (S)	%						92	97	50-200			
13C3HFPO-DA(S)	%						85	86	50-200			
13C4-PFBA (S)	%						93	95	50-200			
13C4-PFHpA (S)	%						87	89	50-200			
13C5-PFHxA (S)	%						88	90	50-200			
13C5-PFPeA (S)	%						85	86	50-200			
13C6-PFDA (S)	%						81	82	50-200			
13C7-PFUdA (S)	%						83	81	50-200			

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QUALITY CONTROL DATA

Project: UCMR5_SE2_May2025
Pace Project No.: 35956761

MATRIX SPIKE & MATRIX SPIKE DUPLICATE:													
6036602				6036603									
		35956752001	MS Spike	MSD Spike	MS	MSD	MS	MSD	% Rec		Max		
Parameter	Units	Result	Conc.	Conc.	Result	Result	% Rec	% Rec	Limits	RPD	RPD	Qual	
13C8-PFOA (S)	%						90	93	50-200				
13C8-PFOS (S)	%						91	94	50-200				
13C9-PFNA (S)	%						85	88	50-200				

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REPORT OF LABORATORY ANALYSIS



QUALITY CONTROL DATA

Project: UCMR5_SE2_May2025

Pace Project No.: 35956761

QC Batch:	1100308	Analysis Method:	EPA 537.1
QC Batch Method:	EPA 537.1	Analysis Description:	537.1 PFOA Compounds, UCMR
		Laboratory:	Pace Analytical Services - Ormond Beach

Associated Lab Samples: 35956761001, 35956761003

METHOD BLANK: 6036470 Matrix: Water

Associated Lab Samples: 35956761001, 35956761003

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
NEtFOSAA	ug/L	0.0017 U	0.0060	0.0017	06/13/25 17:31	
NMeFOSAA	ug/L	0.0020 U	0.0060	0.0020	06/13/25 17:31	
PFTeDA	ug/L	0.0027 U	0.0080	0.0027	06/13/25 17:31	
PFTrDA	ug/L	0.0024 U	0.0070	0.0024	06/13/25 17:31	
13C2-PFDA (S)	%	99	70-130		06/13/25 17:31	
13C2-PFHxA (S)	%	97	70-130		06/13/25 17:31	
HFPO-DAS (S)	%	95	70-130		06/13/25 17:31	
NEtFOSAA-d5 (S)	%	97	70-130		06/13/25 17:31	

LABORATORY CONTROL SAMPLE: 6036471

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
NEtFOSAA	ug/L	0.005	0.0044 I	89	50-150	
NMeFOSAA	ug/L	0.006	0.0055 I	91	50-150	
PFTeDA	ug/L	0.008	0.0073 I	92	50-150	
PFTrDA	ug/L	0.007	0.0065 I	92	50-150	
13C2-PFDA (S)	%			92	70-130	
13C2-PFHxA (S)	%			89	70-130	
HFPO-DAS (S)	%			85	70-130	
NEtFOSAA-d5 (S)	%			93	70-130	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 6036472 6036473

Parameter	Units	35956752001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
NEtFOSAA	ug/L	0.0016 U	0.038	0.037	0.035	0.035	93	93	70-130	1	30	
NMeFOSAA	ug/L	0.0019 U	0.045	0.045	0.042	0.041	92	92	70-130	1	30	
PFTeDA	ug/L	0.0025 U	0.06	0.06	0.057	0.057	94	96	70-130	0	30	
PFTrDA	ug/L	0.0022 U	0.053	0.052	0.050	0.050	94	95	70-130	0	30	
13C2-PFDA (S)	%						101	102	70-130			
13C2-PFHxA (S)	%						90	91	70-130			
HFPO-DAS (S)	%						84	86	70-130			
NEtFOSAA-d5 (S)	%						97	97	70-130			

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QUALIFIERS

Project: UCMR5_SE2_May2025

Pace Project No.: 35956761

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.

ND - Not Detected at or above adjusted reporting limit.

TNTC - Too Numerous To Count

MDL - Adjusted Method Detection Limit.

PQL - Practical Quantitation Limit.

RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.

S - Surrogate

1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.

Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.

LCS(D) - Laboratory Control Sample (Duplicate)

MS(D) - Matrix Spike (Duplicate)

DUP - Sample Duplicate

RPD - Relative Percent Difference

NC - Not Calculable.

SG - Silica Gel - Clean-Up

U - Indicates the compound was analyzed for, but not detected.

N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.

Reported results are not rounded until the final step prior to reporting. Therefore, calculated parameters that are typically reported as "Total" may vary slightly from the sum of the reported component parameters.

Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.

TNI - The NELAC Institute.

ANALYTE QUALIFIERS

I The reported value is between the laboratory method detection limit and the laboratory practical quantitation limit.

U Compound was analyzed for but not detected.

N2 The lab does not hold NELAC/TNI accreditation for this parameter but other accreditations/certifications may apply. A complete list of accreditations/certifications is available upon request.

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: UCMR5_SE2_May2025

Pace Project No.: 35956761

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
35956761001	10413 Crosby Wellfield Service	EPA 200.7	1102849	EPA 200.7	1102958
35956761003	12269 44 Road Water Service Bu	EPA 200.7	1102849	EPA 200.7	1102958
35956761001	10413 Crosby Wellfield Service	EPA 533	1100351	EPA 533	1100573
35956761003	12269 44 Road Water Service Bu	EPA 533	1100351	EPA 533	1100573
35956761001	10413 Crosby Wellfield Service	EPA 537.1	1100308	EPA 537.1	1100526
35956761003	12269 44 Road Water Service Bu	EPA 537.1	1100308	EPA 537.1	1100526

REPORT OF LABORATORY ANALYSIS

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Project #
Project Manager:
Client:

WO#: 35956761

PM: RJS

Due Date: 06/25/25

CLIENT: CADIMIUCMR4

Courier: ☒ Fed Ex ☐ UPS ☐ USPS ☐ Client ☐ Commercial ☐ Pace

Tracking # 4348 2255 1442

☐ Other:

Custody Seal: ☒ Yes ☐ No

Seal intact: ☒ Yes ☐ No

Packing Material: ☐ Bubble Wrap

☐ Bubble Bags

☒ None ☐ Other

Ice: ☒ Yes ☐ No

	Yes	No	Comments
Chain of Custody Present	☒	☐	
Chain of Custody Completed	☒	☐	
Relinquished & Sampler Signature	☒	☐	
Samples Arrived within Hold Time	☒	☐	
Sufficient Volume	☒	☐	
Correct Containers Used	☒	☐	
Containers Intact	☒	☐	
Sample Labels match COC	☒	☐	
Samples received ≤48 hr from collection?	☒	☐	
If no, were samples kept ≤6°C after 48 hrs of collection?	☒	☐	
Person contacted for verification	☐	☐	

Temperature Verification

IR Gun ID: +409

Initials: SB4

Date: 5/20/25

Time: 1123

CF (°C): 0

	Observed (°C)	Corrected (°C)	Comments
EPA 533	2.0	2.0	
EPA 537.1	2.3	2.3	

Preservation Verification

pH strip: 4.6441704

Free Cl Strip/DPD: 2502171

pH meter: SR-3

Cl Meter: SR-1

200.7 Lab Preserved: Nitric Acid:

Date/Time:

Initials:

Sample/Container ID	pH				Free Chlorine**		Comments
	533* (pH 6.0-8.0)	537.1 (pH 6.0-8.0)	200.7 (pH <2)	200.7 Lab Preserved	533	537.1	
10413 Crosby wellfe. 1/4	7.3	7.5	2		ND	ND	
11 2/4	7.3	7.5			ND	ND	
11 3/4	7.4	7.5			ND	ND	
11 4/4	7.4	7.5			ND	ND	
Feb-10413 Crosby...	6.7	7.5			ND	ND	
12269441 Road water 1/3	6.7	7.5	2		ND	ND	
11 2/3	7.0	7.5			ND	ND	
11 3/3	7.2	-			ND	-	

*533 pH may be adjusted at the bench prior to extraction.

**ND = <0.1 mg/L

Labeled by: SB4

Reviewed by: EAS

Delivered by:

Preservation Verification Continued

[illegible]

*533 pH may be adjusted at the bench prior to extraction ** ND = <0.1 mg/L